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EXP.NO:3 Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

CODE:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain',
               'Rain', 'Overcast', 'Sunny', 'Sunny', 'Rain',
               'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool',
                   'Cool', 'Cool', 'Mild', 'Cool', 'Mild',
                   'Mild', 'Mild', 'Hot', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal',
                'Normal', 'Normal', 'High', 'Normal', 'Normal',
                'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes',
             'No', 'Yes', 'No', 'Yes', 'Yes',
             'Yes', 'Yes', 'Yes', 'No']
}
df = pd.DataFrame(data)
print("Training Dataset:\n")
print(df)
le = LabelEncoder()
encoded_df = df.copy()
for column in encoded_df.columns:
    encoded_df[column] = le.fit_transform(encoded_df[column])
X = encoded_df.drop('Play', axis=1)
y = encoded_df['Play']
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)
print("\nDecision Tree:\n")
print(export_text(model, feature_names=list(X.columns)))
new_sample = pd.DataFrame({
    'Outlook': ['Sunny'],
    'Temperature': ['Cool'],
    'Humidity': ['High'],
    'Wind': ['Strong']
})
encoded_sample = pd.DataFrame()
for column in new_sample.columns:
    encoder = LabelEncoder()
    encoder.fit(df[column])
    encoded_sample[column] = encoder.transform(new_sample[column])

prediction = model.predict(encoded_sample)
play_encoder = LabelEncoder()
play_encoder.fit(df['Play'])
result = play_encoder.inverse_transform(prediction)
print("\nNew Sample:")
print(new_sample)
print("\nPrediction:")
print(result[0])
OUTPUT:
```

Training Dataset:

	Outlook	Temperature	Humidity	Wind	Play
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

Decision Tree:

```
|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook > 0.50
|   |--- Humidity <= 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- Wind <= 0.50
|   |   |   |   |--- class: 0
|   |   |   |   |--- Wind > 0.50
|   |   |   |       |--- class: 1
|   |   |--- Outlook > 1.50
|   |   |   |--- class: 0
|   |--- Humidity > 0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Temperature <= 1.00
|   |   |   |   |--- class: 0
|   |   |   |   |--- Temperature > 1.00
|   |   |   |       |--- class: 1
|   |   |--- Wind > 0.50
|   |       |--- class: 1
```

New Sample:

	Outlook	Temperature	Humidity	Wind
0	Sunny	Cool	High	Strong

Prediction:

No